

Smart Parking Slot Monitor Using Ultrasonic Sensor

Abstract

This project presents a Smart Parking Slot Monitoring System using an HC-SR04 ultrasonic sensor and Arduino Uno. The system detects whether a parking slot is occupied or vacant by measuring the distance between the sensor and the object. The collected data is analyzed using Python to understand parking usage patterns and identify peak demand times. This low-cost and efficient solution helps in reducing parking search time and improving parking management.

1. Introduction

Parking management is a major challenge in urban areas due to the increasing number of vehicles. Traditional parking systems require manual monitoring, which is inefficient and time-consuming. Smart parking systems use sensors and microcontrollers to automatically detect parking availability. This project demonstrates a simple yet effective smart parking slot monitor using IoT concepts.

2. Problem Statement

To design and implement a system that can detect parking occupancy using an ultrasonic sensor and analyze the collected data to understand parking usage trends.

3. Objectives

- Detect whether a parking slot is occupied or empty
 - Control a servo motor to simulate a parking gate
 - Indicate parking status using LEDs
 - Collect parking data over time
 - Analyze data using Python for occupancy intervals and daily plots
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4. Hardware Components

- Arduino Uno
- HC-SR04 Ultrasonic Sensor
- Servo Motor (SG90)
- LEDs (Red and Green)
- Resistors (220Ω)

- Breadboard
 - Jumper Wires
 - USB Cable / Power Supply
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5. Software Tools

- Arduino IDE
 - Tinkercad (for simulation)
 - Python
 - Libraries: Pandas, Matplotlib
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6. Circuit Description

The HC-SR04 ultrasonic sensor is connected to the Arduino using digital pins for trigger and echo. The servo motor is connected to a PWM pin to simulate a parking gate. LEDs are connected to digital pins through current-limiting resistors to indicate slot status. All components share a common ground.

7. Working Methodology

1. The ultrasonic sensor sends sound waves using the trigger pin.
 2. The echo pin receives the reflected waves from an object.
 3. Arduino calculates the distance based on the time delay.
 4. If the distance is below a threshold, the slot is considered occupied.
 5. The servo motor rotates to open the gate when a vehicle is detected.
 6. LEDs indicate whether the slot is occupied or free.
 7. Distance and occupancy data is logged.
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8. Data Collection Plan

- Distance readings are recorded at regular intervals.
 - Each reading is converted into an occupied or vacant status.
 - The data is stored in a CSV file for analysis.
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9. Python Data Analysis

Python is used to analyze the collected parking data. - Occupancy intervals are calculated - Daily occupancy graphs are plotted - Peak usage times are identified

These analyses help in understanding parking demand patterns.

10. Results and Discussion

The system successfully detects parking occupancy with good accuracy. The servo motor responds correctly to vehicle presence. Python analysis shows clear peak hours, indicating higher parking demand during specific times of the day.

11. Conclusion

The Smart Parking Slot Monitoring System effectively integrates hardware and software to provide an automated parking solution. It reduces human intervention and provides useful data insights through analysis. The project achieves all its objectives successfully.

12. Future Work

- Vehicle type identification using camera and AI
 - Cloud-based data storage
 - Mobile application for real-time parking status
 - Multiple slot monitoring
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13. References

- Arduino Official Documentation
- HC-SR04 Sensor Datasheet
- Python Data Analysis Libraries Documentation